

Rebuttal: IndicAlignProbe: Measuring Language-Conditioned Safety Asymmetry in Hinglish LLM Interactions

Reviewer Scores Summary

Reviewer	Soundness	Confidence	Overall Assessment
R1	3.5	3	4
R2	3	3	2
R3	3	4	2

Response to Reviewer 1

We thank Reviewer 1 for their careful reading and constructive feedback. We are grateful that the reviewer found the paper well written, recognized the multi-axis equivalence auditor as a meaningful contribution, appreciated the mathematical formalization of the flip-rate framework, and acknowledged the human-annotation validation of our judge. Below we respond to each concern in turn and describe the concrete revisions we will make. Several of the requested revisions also directly address the reviewer's lower scores on Reproducibility, Datasets, and Software, and we summarize our release plan in Point 6.

Point 1: Definitions of the three jailbreak strategies and the absence of role prompting

The three strategies like symbolic masking, scenario nesting, and logical appeal are not defined in the paper properly; it is assumed that the reader is well familiar with the research stream on symbolic encoding for jailbreaks; further, it is unclear why other jailbreaking strategies like role prompting are not considered.

We agree that the strategies are used before they are formally defined, and we will fix this. In the revision we will add a dedicated paragraph in Section 4.1 (or a short Section 4.6) that defines each strategy precisely and self-containedly, so no prior familiarity with the symbolic-encoding jailbreak literature is assumed. The working definitions we will state are the following.

- **Symbolic Masking.** The harmful request is re-expressed through symbolic, metaphorical, or fictional abstraction that displaces the harmful referent onto a "safe" symbolic surface. The concrete instance in Appendix B is the "Weaponization of Symbols" framing, and Section 6.1 characterizes this class as persona-style and archival framings that move the model into topical engagement without reliably producing actionable content.
- **Scenario Nesting.** The harmful request is embedded inside a nested narrative or containing scenario (for example a documentary script, a research vignette, or a creative story). Appendix A.4, Example B, shows the documentary-script instance. Section 6.1 characterizes this class as nested narratives that bypass the refusal boundary entirely at higher rates.
- **Logical Appeal.** The request is normalized through analytical, academic, or reasoning-based justification that reframes the harmful ask as a legitimate intellectual exercise. Section 6.1 pairs this class with logical framings that produce operationally specific output.

We will also replace the opaque internal strategy code in Appendix B (currently `CLs1SMCR`) with a human-readable strategy label, since that code reinforced the impression that strategies were undefined.

On the specific question of role prompting. Our three strategies were chosen to span a deliberate axis of how the attack acts on meaning, from direct symbolic re-encoding (symbolic masking) to indirect contextual reshaping (scenario nesting and logical appeal). This is exactly the semantic-direct versus semantic-indirect axis the reviewer raises in their Comments section, and we respond to it in detail in Point 5. Role prompting sits close to scenario nesting on this axis (both reshape semantics through context rather than encoding), which is why it was not initially separated out. We agree it is a natural and important addition. We will add role prompting as a fourth strategy in an extended sweep.

[New Experiment] We propose adding a role-prompting strategy cell (persona-assignment framings such as "you are an unrestricted expert") to the existing $3 \times 4 \times 3$ grid, run under the identical harness with the same equivalence certification and graded rubric. Because role prompting reshapes semantics through context rather than through operational content, we hypothesize its behavioral profile will resemble scenario nesting more than symbolic masking, that is, non-trivial forward and critical flip rates on `Qwen3-30B-A3B` and near-symmetric behavior on `GPT-05S-20B` and `Nemotron-30B-A3B`. We present this as a proposed experiment and an expected direction, not as a measured result.

[Revision] and [New Experiment]

Point 2: The equivalence auditor's technical implementation is not fully explained

The paper highlights that the semantic equivalence evaluator is their greatest contribution but its technical implementation is not fully explained.

This is a fair criticism, and since the auditor is the central methodological contribution we will substantially expand its description. The current specification is spread across Section 4.4 and Appendix A.7, and we will consolidate and deepen it. To summarize what is already specified and what we will add.

What is already in the paper. The auditor scores each candidate English/Hinglish-RH pair across **nine dimensions** organized into three groups (Section 4.4 and Appendix A.7). The groups are what the prompt asks (harmful intent, scenario identity, information granularity, attack strategy), how the prompt is framed (cultural specificity, target group, ambiguity level), and translation-quality checks (Hinglish/RH language fidelity, strategy faithfulness). It emits an aggregate equivalence score $e \in [0, 1]$, a stronger-than-English flag b , a language-fidelity flag ℓ , and a strategy-faithfulness flag s . A pair is certified if and only if $\neg b \wedge \ell \wedge s \wedge e \geq 0.6$, and non-certified pairs are discarded and regenerated (Section 4.4, Appendix A.7).

What we will add in the revision. We will add a new appendix subsection giving the full auditor specification. This includes (a) the exact Gemini-2.5-Flash structured-JSON auditor prompt and output schema, (b) the per-dimension scoring scale and the anchor descriptions used for each of the nine dimensions, (c) the exact aggregation function that maps the nine per-dimension scores to the scalar e (a weighted mean over the three groups with the content group weighted highest), and (d) the precise logical conditions under which the binary flags b , ℓ , and s are set. We will also report auditor behavior statistics over the full sweep, namely the fraction of generated pairs rejected at certification and the marginal rejection rate attributable to each flag, so a reader can see how selective each gate is. Finally we will release the auditor prompt and code (see Point 6), which makes the auditor fully reproducible rather than described only in prose.

[Revision]

Point 3: The weights in Equation 4 are not explained

The weights in equation 4 are not explained; why did the authors choose 0.55 etc.

We will explain these weights explicitly, and we want to first clarify an important scoping point that we did not make prominent enough in the current text.

Equation 4,

$$F(g) = 0.55 \cdot \mathbf{1}[\text{flip}] + 0.20 \cdot e \cdot \mathbf{1}[\neg b] + 0.10 \cdot \mathbf{1}[c_{\text{EN}}] + 0.10 \cdot \mathbf{1}[c_{\text{RH}}] + 0.05 \cdot \text{nat}(p_{\text{RH}}),$$

is the objective used only by the GEPA generator during the prompt-generation optimization phase, which was frozen before the main 1,512-pair evaluation sweep (Appendix A.6). It does not enter the evaluation metrics, and it does not relax the certification requirement. Every pair that reaches a target model must still pass the hard gate $\neg b \wedge \ell \wedge s \wedge e \geq 0.6$ from Section 4.4. So the soft weight $0.20 \cdot e$ in $F(g)$ steers the generator toward more equivalent proposals during

search, but final equivalence validity is enforced by the separate hard certification gate, not by this weight. We will state this separation clearly in the revision.

On the specific values, the ordering encodes the priorities of a measurement instrument rather than an attack optimizer (Section 4.3, "measurement informativeness"). Flip discovery receives the dominant weight (0.55) because surfacing behavioral flips is the measurement target of the probe. Equivalence quality is second (0.20) because it is the validity constraint that makes any surfaced flip attributable to language. Judge consistency on each side (0.10 each for c_{EN} and c_{RH}) and Hinglish/RH naturalness (0.05) act as regularizers that keep proposals well-formed and gradeable without dominating the search. We will add this rationale to Section 4.3.

[New Experiment] To show the findings do not depend on the specific values, we propose a generator-objective sensitivity ablation. We will re-run the GEPA generator under several perturbed weight vectors (for example shifting mass between the flip term and the equivalence term across a small grid), regenerate certified pairs, and re-measure the headline quantities, namely the per-model refusal gaps Δ in Table 1 and the strategy-level critical-flip ordering in Figure 2. Because all pair sets pass the same hard certification gate regardless of the generator weights, we hypothesize the qualitative conclusions (Qwen3-30B-A3B asymmetrically unsafe, the other two near-symmetric, and symbolic masking producing much lower actionable harm than the other two strategies) will be stable across these settings. We present this as a proposed robustness check and an expected direction, not as measured numbers.

[Clarification] and [New Experiment]

Point 4: A larger human evaluation of the LLM-as-a-judge would be appreciated (about 10 percent)

A larger sample of prompts/result pairs by human evaluation of the LLM-as-a-judge would be appreciated; about ~10% subsample would be appreciated.

We agree and will expand the human validation. Our current audit covers 100 prompt-response pairs stratified across models, harm categories, and strategy classes, which is roughly 6.6 percent of the 1,512 pairs, and it yields 86 percent exact agreement, 99 percent adjacent agreement, Cohen's $\kappa = 0.79$, and quadratic weighted $\kappa = 0.94$ (Appendix A.5). The reviewer's request for a roughly 10 percent subsample corresponds to about 151 pairs.

[New Experiment] We will expand the human audit to at least 160 pairs (above 10 percent), keeping the stratification across the model $\times$ category $\times$ strategy cells so that agreement is measured evenly across the grid rather than concentrated in a few cells. We will also add a second independent annotator on the expanded set and report inter-annotator agreement alongside human-judge agreement, which lets a reader separate judge error from intrinsic rubric ambiguity. Given that most current disagreements occur at adjacent boundaries and that the automated judge tends to score more conservatively than humans (Appendix A.5), we expect

agreement on the larger sample to remain in a comparable range (Cohen's κ above 0.7 and quadratic weighted κ near or above 0.9). We present these as expected outcomes of the proposed expanded audit, not as measured values.

We would also like to draw the reviewer's attention to an orthogonal validation that does not rely on the LLM judge at all. Appendix A.1 and Figure 3 report a fully judge-independent directional signal based only on response length (short English refusal paired with long Hinglish/RH engagement). This signal independently reproduces the headline asymmetry, showing a 152:0 forward-to-reverse ratio for Qwen3-30B-A3B and no directional preference for the two baseline models. Because it uses no semantic judgment, it corroborates the judge-based findings through a mechanism that cannot inherit judge error. We will make this cross-validation more prominent in the revision.

[Acknowledged] and [New Experiment]

Point 5: Consider strategies that tackle the semantic layer directly (encoding) versus indirectly (context)

Since semantic consistency evaluation is an essential focus of the paper, it would be useful to consider additional strategies that tackle the semantics directly, for example encoding, versus strategies that shape the semantics of the problem indirectly (e.g. through context).

We find this a genuinely useful framing and we will adopt the direct-versus-indirect axis explicitly in the revision. It also connects tightly to the premise of the paper, since Hinglish/Romanized Hindi is itself a surface-level transformation of the semantic content (transliteration and code-switching), which is precisely a semantic-direct manipulation of the prompt's realization. In that sense the whole probe already studies one semantic-direct axis (script and register), while holding meaning fixed through the equivalence auditor.

Our three current strategies partially span the reviewer's axis. Symbolic masking operates more directly on the semantic surface by re-encoding the harmful referent, while scenario nesting and logical appeal reshape meaning indirectly through surrounding context. We will make this taxonomy explicit and use it to organize the strategy discussion in Section 6.1.

[New Experiment] We propose adding an explicitly semantic-direct encoding strategy family, distinct from symbolic masking's metaphorical masking, that manipulates the token-level realization of the request while the equivalence auditor certifies that the underlying meaning is preserved. Candidate members include cipher or substitution encodings, leetspeak, and structured obfuscation of key harmful terms, applied within the Hinglish/RH register. This lets us contrast three levels on the reviewer's axis, namely pure surface encoding, symbolic or metaphorical masking, and context-based framing (nesting, logical appeal, role prompting from Point 1). We hypothesize that semantic-direct encoding will, like symbolic masking, more often produce topical engagement than fully actionable output, whereas context-based strategies will

more often cross into actionable harm, sharpening the qualitative distinction we already report in Section 6.1 and Figure 2. We present this as a proposed extension and an expected direction, not as measured results.

[Revision] and [New Experiment]

Point 6: Reproducibility, code, and data release

(Implicit in the Reproducibility, Datasets, and Software scores.)

We take these scores seriously and will make a full artifact release, which we should have committed to explicitly in the submission. Concretely we will release, under a gated research-use license appropriate for safety-sensitive material, the following.

- The full evaluation harness, including the orchestration built on the Hermes-Agent framework, JSON-contract enforcement, retrieval gating, and DSPy/GEPA integration described in Section 4.3.
- All 1,512 certified English/Hinglish-RH prompt pairs together with their per-pair metadata (harm category, strategy, scenario metadata, information granularity, target group, and the auditor's per-dimension equivalence scores and flags).
- The multi-axis equivalence auditor, including its exact prompt, output schema, per-dimension scoring anchors, and aggregation code (this directly supports the expansion requested in Point 2).
- The graded four-level compliance judge implementing the three-gate rubric of Equation 5, and the GEPA-optimized generator seed and objective from Equation 4 and Appendix A.6.
- The human annotation data from the judge-validation study, including the expanded audit described in Point 4, with the per-item human and automated scores so the agreement statistics in Appendix A.5 can be recomputed independently.
- The evaluation and analysis scripts that regenerate Table 1, Figures 2 and 3, and the appendix breakdowns.

Because the entire pipeline is automated end-to-end (Section 4.1) and all target models are open-weight and evaluated at temperature $T = 0$ with empty system prompts (Section 5.2), a reader with the released artifacts can reproduce the certified pair set, the target-model responses, and the reported metrics. We will add a reproducibility appendix documenting exact model versions, decoding settings, and run instructions, and we will note the responsible-release considerations given the safety-sensitive content.

[Acknowledged] and [Revision]

We thank the reviewer again for feedback that will make the strategy taxonomy, the auditor, and the reproducibility of the work substantially clearer. We believe the additions above (explicit

strategy definitions, a role-prompting cell, an encoding-based semantic-direct strategy, a full auditor specification, an Equation 4 clarification with a sensitivity ablation, an expanded human audit above 10 percent, and a complete artifact release) directly resolve the concerns raised.

Response to Reviewer 2

We thank Reviewer 2 for their careful reading and constructive feedback. We are grateful that the reviewer found the motivation clear, the alignment-comprehension decoupling formalization a decent formal attempt, and the ablations interesting. Below we address every concern point by point. We group the presentation concerns first (Points 1 to 3) and then respond in depth to the central novelty concern (Points 4 to 6), which we take most seriously.

Point 1: The name "IndicAlignProbe" suggests broader coverage than Hinglish

First of all the name is misleading. The authors only audit in Hinglish, but as the authors acknowledge in the limitations and future work, there are other Indic languages.

We agree that the "Indic" prefix can imply broader linguistic coverage than the single Hinglish/Romanized Hindi instantiation we evaluate, and we thank the reviewer for flagging this. We want to clarify the intended scope and then commit to concrete fixes.

The framework itself is language-agnostic by construction. The three load-bearing components, the multi-axis equivalence auditor (Section 4.4), the three-gate graded compliance rubric (Section 4.5, Equation 5), and the GEPA-based measurement harness (Section 4.3), do not encode anything Hinglish-specific except the language-fidelity axis ℓ and the naturalness heuristic $\text{nat}(p_{RH})$. Swapping the target register to Bengali, Marathi, Tamil, or Urdu requires changing only those two language-specific checks, which is exactly the extension we describe in the Limitations and Conclusion. In that sense "Indic" reflects the designed generality of the measurement primitive, while Hinglish is the first instantiated case study.

That said, the reviewer is right that the current framing under-signals this. We will make the following revisions.

1. We will retitle to make the instantiation explicit, for example "IndicAlignProbe: An Equivalence-Certified Probe for Language-Conditioned Refusal Asymmetry, Instantiated for Hinglish." The current subtitle already says "in Hinglish," and we will carry that scoping into the system name framing so the abstract and Section 1 state plainly that Hinglish/RH is the single evaluated register and that other Indic registers are a stated extension, not a claim of the present paper.
2. We will add one sentence to the abstract and to the Contributions list stating that the harness is language-agnostic apart from the two language-specific checks, so the reader is not left to

infer scope from the name.

[Revision]

Point 2: Forward and reverse flips appear before they are defined

In the introduction, the authors suddenly introduce forward and backward flips, but they are explained further late in the paper.

We agree this hurts readability, and we will fix it. The formal definitions currently live in Section 3 (forward flip, $s_{EN} = 0 \wedge s_{RH} \geq 2$; critical flip, $s_{EN} \leq 1 \wedge s_{RH} = 3$) and the reverse-flip direction is only made precise via Equation 2 and the directional ratio in Section 5.3. A reader meeting the terms earlier has no anchor.

We will add a one-line intuitive gloss at the first mention, before any formal notation. Concretely, we will insert wording of the form "A forward flip is a pair where the model refuses the English prompt but produces partial or fully actionable harmful content for the equivalent Hinglish/RH prompt. A reverse flip is the opposite direction, where Hinglish/RH is refused but English complies." We will attach an explicit forward reference to Section 3 for the formal definitions and to Figure 1 for the pipeline picture. We will also standardize terminology, since the abstract currently says "informative behavioral flips" while later sections say "forward/reverse/critical," and we will use the same three named terms consistently from first use.

[Revision]

Point 3: The problem formulation should be presented more clearly

The authors need to present the problem formulation in an even better way, as this should be a part of the paper that helps the readers easily understand the main contribution of the paper. But in its current version, it fails to do so.

This is a fair criticism and we will restructure Section 3 for clarity. Our planned revision does the following.

1. Lead with intuition before formalism. We will open Section 3 with a concrete worked pair (an English prompt that is refused next to its certified-equivalent Hinglish/RH prompt that is answered), drawn from Appendix B, so the reader sees the phenomenon before meeting $\mathcal{C}_\theta$, $\mathcal{S}_\theta$, and the flip predicates.
2. Add a compact notation table. We will add a small glossary mapping each symbol to a plain-language meaning ($\mathcal{C}_\theta$ comprehension, $\mathcal{S}_\theta$ refusal gate, J the four-level score, forward/critical flips, Δ_c, ρ), so the reader does not have to reconstruct notation from prose.
3. State the central claim as a single sentence up front. The core testable prediction is the directional asymmetry in Equation 2, that comprehension transfers across the English to Hinglish/RH shift more reliably than the refusal boundary does, so forward flips strictly

exceed reverse flips. We will foreground this one sentence as the thesis of the formulation, then present the measurement machinery as the way we test it.

4. Tighten the link to the contribution. We will make explicit that the formulation exists to license a causal reading, that a measured gap $\Delta_c(m)$ is attributable to language rather than to prompt drift because equivalence is certified (Equation 1) before any model is queried.

We believe these changes directly address the reviewer's concern that the formulation currently under-serves the reader's understanding of the contribution.

[Revision]

Point 4: The lack of transfer is well-known intuitive wisdom, which questions the contribution

My second foremost concern is that this lack of transfer appears to be a well-known intuitive wisdom to the LLM-alignment community. Training on iid prompts, it is expected that it won't transfer robustly to ood prompts ... This questions the contribution of the paper, beyond the proposed auditing approach specific to a single indic language given the insights are well-known wisdom.

We appreciate this challenge and want to respond substantively, because we agree that "safety may transfer imperfectly out of distribution" is a widely held intuition. Our position is that the contribution of the paper is not the existence of some gap. It is converting a folk intuition into a controlled, causally-attributed, severity-graded, and model-resolved measurement, and in doing so surfacing three specific findings that the folk intuition does not predict and in two of three cases actually contradicts. We take the reviewer's own analogy (train on task A, expect weak generalization to task B) and show where our results depart from it.

(a) The folk intuition conflates confounds that our design removes, which is the methodological contribution, not a restatement. The intuition "OOD prompts generalize worse" says nothing about why a specific multilingual prompt failed. As we note in Section 1, when a Hinglish prompt elicits unsafe behavior that its English counterpart did not, the gap "could reflect language, framing, specificity, ambiguity, or adversarial structure, and prior work cannot tell us which." Existing multilingual jailbreak evaluations vary all of these simultaneously through translation, so they cannot attribute the gap to language. Our multi-axis equivalence auditor (Section 4.4, nine dimensions across intent, framing, and translation fidelity, certified iff $\neg b \wedge \ell \wedge s \wedge e \geq 0.6$) certifies that the English and Hinglish/RH prompts match on harmful intent, scenario, granularity, strategy, cultural specificity, target group, and ambiguity before the model is ever queried. This is precisely what makes the residual gap $\Delta_c(m)$ attributable to language as the sole intended variable (Equation 3, Figure 1). The intuition tells you a gap might exist. It does not, and cannot, tell you that a measured gap is caused by language rather than by prompt drift. Providing that causal control is a contribution, and the reviewer's phrase "beyond the proposed

auditing approach" concedes the auditing approach itself. We would argue the auditing primitive is not "specific to a single Indic language" at all, as discussed in Point 1 it is language-agnostic apart from two checks.

(b) The graded-severity finding contradicts a monotone reading of the intuition and is invisible to binary metrics. The "better coverage, better generalization" intuition predicts a roughly monotone degradation in refusal rate. It does not predict that failure severity dissociates from failure frequency by strategy. We find exactly that dissociation (Section 6.1). Symbolic masking produces topical engagement at a rate comparable to the other strategies (78 forward flips, 15.5%) yet crosses into fully actionable harm at a $\geq 15\times$ lower rate than scenario nesting and logical appeal (critical-flip rate 1.0% versus 15.1% and 9.1%). Two strategies with similar forward-flip rates therefore carry very different real-world threat. Under a binary attack-success metric these look comparable. Only the four-level rubric (Gates A , B , C in Equation 5, separating topical acknowledgement from operational specificity from actionability) exposes the distinction. The folk intuition offers no purchase on which failures are dangerous versus merely on-topic, and building the instrument that answers that question is a contribution.

(c) The model-specificity finding falsifies the universal reading of the intuition. If "English-centric post-training does not transfer to OOD registers" were a blanket law, every English-centric model should show the gap. Under one identical harness we instead observe three qualitatively distinct regimes (Table 1, Section 6.2). Qwen3-30B-A3B is Asymmetrically Unsafe, with a +35.9 pp English/Hinglig gap and a strict 152:0 forward-to-reverse directional ratio (Figure 3). GPT-OSS-20B is Symmetrically Safe (+3.6 pp). Nemotron-30B-A3B is Symmetrically Permissive (+2.6 pp). Two of three models show essentially no language-conditioned asymmetry. This is an existence result, not a universality claim (we state this explicitly in Limitations), and it directly contradicts the reading of the reviewer's intuition that would predict degradation everywhere. It also matters practically, because it is a constructive proof that the gap is not inherent to the register. Some models already close it, so it is a fixable engineering target rather than an inevitability. The same-harness design is what licenses this conclusion, since if the harness merely inflated failures indiscriminately all three targets would fail, and they do not.

(d) The phenomenon is a decoupling over the same task, not degradation on a different task. The reviewer's analogy frames English and Hinglish as task A and task B. Our certified pairs are, by construction, the same task, the same harmful intent and operational content, differing only in surface register (Equation 1). The finding is subtler than "performance drops on a new task." It is that two capabilities dissociate over identical semantic input. Comprehension clearly transfers, because a forward flip requires the Hinglish/RH response to be topically on point and operationally specific (Gates A and B satisfied, Section 3 "Inferring Comprehension from Response Content"), yet the refusal boundary that fired for the equivalent English prompt did not fire. Folk wisdom about coverage does not tell you which of a model's capabilities survives the shift and which does not. Our forward-flip construction pins that down, comprehension present

exactly where refusal is absent, and Equation 2 turns it into a directional prediction (forward flips strictly exceed reverse flips) that we then confirm at a strict 152:0 ratio in Qwen3 with a judge-independent length signal (Figure 3).

(e) Measurement is the scientific contribution. A widely believed intuition and a measured phenomenon are not the same object, and the field cannot act on the former. Practitioners serving the hundreds of millions of Hinglish users referenced in Section 1 cannot mitigate "we suspect OOD is worse." They can act on a quantified 35.9 pp gap, a 152:0 directional asymmetry, a $15\times$ severity gap between strategies, and a localization map (Figure 2) showing that failures concentrate in specific model $\times$ category $\times$ strategy cells rather than spreading uniformly. We would gently note that in two of three models the intuition is not merely unquantified but empirically wrong, which is the strongest possible argument that the intuition was not a substitute for the measurement.

For these reasons we respectfully disagree that the insights reduce to well-known wisdom. We will, however, sharpen the framing in Section 1 and the Conclusion to state explicitly what the paper adds beyond the intuition, namely controlled causal attribution, graded severity, and model-resolved regimes, so the delta over prior belief is unmistakable to the reader.

[Rebuttal]

Point 5: Proposed control isolating language from generic distribution shift

To address the reviewer's iid-versus-ood framing head on, we propose a control experiment that we believe directly tests their hypothesis and that we can add.

If the effect were driven purely by generic distribution shift rather than by language, then any sufficiently out-of-distribution English register (rare phrasings, archaic or heavily code-styled English, unusual sociolects) should degrade refusal by a comparable amount. We propose to construct a matched set of certified-equivalent English-to-English pairs where the second element is an unusual but still English register produced and certified by the same auditor, and to compare the refusal gap and directional ratio against the English-to-Hinglig/RH pairs on the same models.

Proposed prediction. We expect the Hinglish/RH shift to produce a substantially larger and directionally asymmetric gap on Qwen3-30B-A3B than the English register shift does, and we expect the two symmetric models to remain near-symmetric under both. If confirmed, this would show the effect is language-conditioned rather than a generic artifact of any OOD register, isolating exactly the variable the reviewer's intuition leaves ambiguous. We present this as a proposal and do not report measured numbers here.

[New Experiment]

Point 6: Proposed ablation demonstrating the causal role of equivalence certification

To make the value of the equivalence auditor empirically visible rather than argued, we propose a certification-ablation. We will take the same seed pairs and evaluate two conditions on identical target models, one using only auditor-certified pairs (as in the paper) and one using uncertified pairs that are allowed to drift (equivalence gate disabled).

Proposed prediction. We expect the uncertified condition to show inflated and less model-specific gaps, because prompt drift reintroduces the framing, specificity, and adversarial-structure confounds that certification removes, whereas the certified condition should preserve the localized three-regime structure of Table 1. A measured separation between the two conditions would demonstrate that certification is doing real causal work and that the reported gaps are not simply generic OOD degradation. This is a proposal, and we state an expected direction rather than fabricated measurements.

[New Experiment]

Point 7: Remaining review items

The reviewer marked "Comments, Suggestions and Typos" as "See Above," so we have treated the writing items in Points 1 to 3. We note the reviewer's Reproducibility (4), Datasets (4), and Software (3) scores and the assessment that limitations and societal impact are handled. We will additionally release the certified 1,512-pair probe set, the auditor and rubric prompts, and the GEPA-optimized generator seed to move the Software rating up and to support independent replication of both the auditing primitive and the three-regime result.

[Acknowledged]

Response to Reviewer 3

We thank Reviewer 3 for their careful reading and constructive feedback. We are grateful that the reviewer recognizes the importance of the multilingual safety-gap problem for code-switched and Romanized registers, the value of controlling semantic equivalence before measuring refusal differences, and the graded four-level rubric that separates topical engagement from partial procedural guidance and fully actionable compliance. We address each of the three weaknesses in turn. Two of them point at experiments we agree would strengthen the paper, and we describe them concretely below as proposals. The third rests on a framing point that we believe we can clarify.

Point 1: Single model family (Gemini-2.5-Flash) across generation, gating, auditing, and judging

Gemini-2.5-Flash is used in several important roles, including relevance gating, prompt generation, auditing, and judging. This may introduce correlated errors or model-specific

bias. The paper would be stronger if generation, auditing, and judging were handled by independent model families.

We agree this is a real concern, and we flag it ourselves in Section 7 (Limitations), where we note that Gemini-2.5-Flash serves multiple orchestration roles and that reported flip rates therefore index harness sensitivity rather than population base rates. We would like to clarify two mechanisms already in the paper that partially decouple the central finding from the single judge, and then propose a direct robustness experiment.

First, the headline result does not depend on any LLM judgment at all. In Section A.1 and §6.2 we compute a fully judge-independent directional signal based only on response length. A pair is forward-shaped when the English response is shorter than 80 characters while the Hinglish/RH response exceeds 500 characters. This metric never calls Gemini. As shown in Figure 3, it reproduces the same three-way separation that the graded judge produces. Qwen3-30B-A3B yields a strict 152:0 forward-to-reverse directional ratio, while GPT-05S-20B and Nemotron-30B-A3B remain non-directional at 59:34 and 42:31. The most consequential claim in the paper, that one model is directionally asymmetric while the other two are not, therefore survives under a signal that shares no model family with the generator or judge.

Second, we validate the judge against human annotation in Section 4.5 and Appendix A.5. On 100 sampled prompt-response pairs the automated judge reached 86% exact agreement and 99% adjacent agreement with human annotators, Cohen's $\kappa = 0.79$, and quadratic weighted $\kappa = 0.94$. We also observe that the automated judge behaves more conservatively than human annotators, so to the extent judge bias exists it tends to under-report compliance rather than inflate the gap.

Third, we note a structural argument against a pure correlated-error explanation. A shared-model bias in generation and judging would most naturally produce a uniform distortion across all three target models, since all three are scored by the identical pipeline under identical conditions. What we observe instead is a sharp separation into three regimes (Table 1) and a strong concentration of severe cells in a single model (Figure 2, where no GPT-05S-20B or Nemotron-30B-A3B cell exceeds 33.3% forward-flip rate). A pipeline artifact that manufactured gaps would have to selectively manufacture a large gap for Qwen3-30B-A3B while leaving the other two near zero under the same harness, which is difficult to reconcile with a generic single-family bias.

That said, the reviewer's point stands that the equivalence auditor and the judge should not both live inside one model family. This is already named in our future-work list in Section 7 and the Conclusion. We propose to elevate it from future work to an added robustness experiment for the revision.

[New Experiment] Cross-judge and cross-auditor robustness. We propose to re-score all 1,512 certified pairs with two additional judges drawn from independent model families, for example a Claude-family model and an open fine-tuned safety judge of the kind used in

HarmBench (Mazeika et al., 2024) and JAILJUDGE (Liu et al., 2024). We will report inter-judge agreement (κ) and test whether (a) the three-regime separation in Table 1 and (b) the directional dominance of Qwen3-30B-A3B persist under each judge. In parallel we will re-audit a random subset of certified pairs with an equivalence auditor from a different family and report certification agreement. As a proposal we expect, but have not yet measured, that the ordinal separation between Qwen3-30B-A3B and the two baselines is robust to judge family, because that separation already reproduces under the judge-free length signal in Figure 3, while absolute flip magnitudes may shift modestly across judges. We will make clear in the text that these are additional-independence checks and report whatever the data show.

[Rebuttal / New Experiment]

Point 2: No direct comparison with established multilingual jailbreak benchmarks or standard safety pipelines

The paper evaluates three models using its own harness, but it does not directly compare with established multilingual jailbreak benchmarks, red-teaming datasets, or standard safety-evaluation pipelines.

We appreciate this suggestion and would like to clarify what IndicAlignProbe is designed to measure, because a naive number-to-number comparison with those benchmarks would compare two different quantities.

Existing multilingual jailbreak benchmarks and red-teaming datasets, such as HarmBench (Mazeika et al., 2024), JailbreakBench (Chao et al., 2024), StrongREJECT (Souly et al., 2024), LinguaSafe (Ning et al., 2025), and IndicJR (Pattnayak and Chowdhuri, 2026a), report aggregate unsafe-completion or attack-success rates over prompts that vary simultaneously in framing, specificity, ambiguity, and adversarial structure. As we argue in the Introduction and Related Work, none of them enforces semantic equivalence between the English member and the non-English member of a pair. Their headline numbers therefore cannot be attributed to language alone, which is precisely the attribution problem IndicAlignProbe was built to solve. Our contribution is a measurement primitive rather than a competing leaderboard, and we state this explicitly in Section 4.3 ("IndicAlignProbe is a probe, not a population estimator") and Section 5.2 (the cross-model comparison "primarily serves as a harness-validity check rather than a leaderboard"). A refusal gap $\Delta_c(m)$ from Equation 3 measures a different object than a benchmark's unsafe-completion rate, so reporting our numbers side by side without a bridge would risk a misleading comparison.

There is a second mismatch. Most of those benchmarks score on a binary attack-success metric, whereas our graded four-level rubric (Equation 5) is the very mechanism that surfaces the symbolic-masking divergence in §6.1, where symbolic masking triggers topical engagement at rates comparable to other strategies but produces actionable harm at a rate at least $15\times$ lower.

Collapsing our severity structure back to binary to match those benchmarks would discard the distinction the reviewer identified as a strength.

We agree, however, that an explicit empirical bridge to external prompts would strengthen the paper, and we propose two concrete additions.

[New Experiment] Re-scoring externally sourced prompts through our pipeline. We propose to take a standard English harmful subset from an established red-teaming set (for example StrongREJECT or the HarmBench behaviors) together with its transliterated or Romanized-Hindi counterparts, or the Romanized-Hindi split of IndicJR (Pattnayak and Chowdhuri, 2026a), and pass these externally authored prompts through our graded compliance judge, and where a matched English counterpart exists, through our equivalence auditor. We will report (a) how our four-level rubric re-scores established prompts relative to their native binary labels and (b) whether the same three-model regime separation appears on prompts we did not author. As a proposal, and not a measured result, we expect the qualitative direction to reproduce, meaning Qwen3-30B-A3B remains the most asymmetric while the other two remain more symmetric, but with different magnitudes because externally sourced prompts are not equivalence-certified. Such a result would demonstrate both interoperability with existing resources and the specific attribution value added by certification.

[Revision] Positioning table. We will add a compact comparison to Related Work that states, for each of these benchmarks and pipelines, what quantity it measures (aggregate unsafe rate versus certified refusal gap), whether it enforces prompt-pair equivalence, and whether it uses binary or graded scoring. This makes the complementary relationship explicit and clarifies why IndicAlignProbe is positioned as a controlled attribution instrument rather than a replacement for breadth-oriented benchmarks.

[Clarification / New Experiment / Revision]

Point 3: Scope, and asymmetry concentrated in a single model

The evaluation focuses only on English-Hinglish/Romanized Hindi and three open-weight models. Also, the strongest asymmetry appears mainly in Qwen3-30B-A3B, while the other two models are more symmetric. The conclusions would be stronger with more Indic languages, scripts, model families, and ablations across harm categories and attack strategies.

We address the three sub-points separately, since one of them is a deliberate finding rather than a limitation.

On asymmetry concentrating in one model. We want to clarify that the concentration of severe behavior in Qwen3-30B-A3B is not a weakness of the result but the core validity check of the methodology, and it is intended. IndicAlignProbe makes an existence claim, not a universality claim, which we state directly in Section 7 (the 35.9 pp asymmetry "support[s] an existence claim

about severe Hinglish/RH decoupling in frontier open-weight models rather than a universality claim"). The key point is that all three models pass through one identical harness and separate into three distinct regimes (Table 1). Qwen3-30B-A3B is Asymmetrically Unsafe at $\Delta = +35.9$ pp, GPT-05S-20B is Symmetrically Safe at $\Delta = +3.6$ pp, and Nemotron-30B-A3B is Symmetrically Permissive at $\Delta = +2.6$ pp. If the pipeline itself induced gaps, all three targets would show large gaps. The fact that two of them show near-zero gaps under the same conditions is exactly what rules out a pipeline artifact and attributes the Qwen3-30B-A3B gap to the model. This interpretation is corroborated by the judge-free directional signal in Figure 3 and by the failure-concentration map in Figure 2, where severe cells localize almost entirely within Qwen3-30B-A3B. We will make this framing more prominent in the abstract and Section 6.2 so that the single-model asymmetry reads as evidence of specificity rather than as a thin result.

On harm-category and attack-strategy ablations. We note that these ablations are already present. The evaluation grid spans four harm categories and three attack strategies (Section 5.1), and the per-cell breakdown in Figure 2, Section 6.1, and Appendix A.3 reports forward and critical flips for every model by category by strategy cell. This is where one of the paper's most novel findings lives, namely the symbolic-masking divergence in §6.1, where symbolic masking induces topical engagement at rates comparable to scenario nesting and logical appeal but yields a critical-flip rate of 1.0% against 15.1% and 9.1% for the other two strategies. We will signpost these ablations more clearly so the reviewer's concern is visibly answered in the main text.

On more languages, scripts, and model families. We agree these extensions would broaden the claim, and the current single-pair, three-model scope was a deliberate choice to establish a clean controlled measurement first. We selected Hinglish/RH because it is the highest-value and hardest case, with hundreds of millions of users, an informal default register, and transliteration variability that makes equivalence certification genuinely difficult, as illustrated in Appendix B. The framework is language-agnostic by construction, since the auditor's language-fidelity axis and the generator instruction can be re-parameterized to other Indic settings. Our future-work list in Section 7 and the Conclusion already names Bengali, Marathi, Tamil, and Urdu. We propose the following extensions.

[New Experiment] Additional model families. We propose to add open-weight models from further families, for example Llama, Mistral, Gemma, and DeepSeek variants, and evaluate them on the identical harness. As a proposal rather than a measured outcome, we expect the three-regime taxonomy to remain populated and at least one additional model to fall into the Asymmetrically Unsafe regime, which would strengthen the existence claim toward a partial prevalence estimate. We will report the regime assignment of every added model regardless of direction.

[New Experiment] Additional languages and scripts. We propose to instantiate the pipeline for at least one additional Indic language and its Romanized script, using the same equivalence auditor with a re-parameterized language-fidelity axis, and to report whether a comparable

directional signal and refusal gap appear. We expect, as a hypothesis, that the effect is not unique to Hinglish/RH, but we will report the measured direction whatever it is.

[Revision] Attack-strategy breadth. We will note in the revised Limitations that the three strategies studied here are not exhaustive, and we will scope a follow-on that adds strategies such as translation-only and multi-turn escalation to test the stability of the symbolic-masking divergence.

[Clarification / New Experiment / Revision]

Summary

The reviewer's three concerns map onto two additive robustness studies and one framing clarification. We will add a cross-judge and cross-auditor robustness experiment with independent model families (Point 1), an empirical bridge that re-scores externally sourced red-teaming prompts through our pipeline plus a positioning table (Point 2), and extensions to more model families, an additional language and script, and broader attack strategies, alongside clearer signposting that the single-model asymmetry and the existing category-by-strategy ablations are deliberate and already present (Point 3). We thank the reviewer again for feedback that we believe makes the paper stronger.

Common Themes and Action Items

Common Themes Across Reviewers

Shared praise for the equivalence auditor and the graded rubric

All three reviewers endorse the two central methodological pieces. Reviewer 1 calls the multi-axis equivalence auditor the paper's central technical contribution and highlights the mathematical formalization of the flip-rate framework. Reviewer 2 credits the alignment-comprehension decoupling formalization and the ablations. Reviewer 3 singles out both the paired-probe equivalence control and the four-level graded rubric that separates topical engagement from partial guidance from fully actionable harm. Our consensus response is to lean into this agreement. We keep the auditor and the rubric as the backbone of the rebuttal and use them to answer the harder concerns, since the graded rubric is what produces the symbolic-masking severity finding and the auditor is what licenses causal attribution.

Scope and generalization (Reviewers 2 and 3, touched by Reviewer 1)

This is the most recurrent theme. Reviewer 2 finds the "Indic" name misleading because only Hinglish/RH is audited. Reviewer 3 notes the evaluation covers only English-Hinglish/RH, only three open-weight models, and that the strongest asymmetry sits in one model. Reviewer 1 asks for additional attack strategies. Our consistent response is twofold. First, we clarify that the framework is language-agnostic by construction and that Hinglish/RH is a deliberate first instantiation chosen as the highest-value and hardest case. Second, we propose concrete breadth extensions (more model families, at least one more Indic language and script, more attack strategies) and we commit to a title and abstract fix that scopes the claim honestly.

The single-model asymmetry is a deliberate finding, not a thin result (Reviewer 3, relevant to Reviewer 2)

Reviewer 3 reads the concentration of asymmetry in Qwen3-30B-A3B as a weakness. This connects to Reviewer 2's novelty concern. Across both rebuttals we make the same argument. The same-harness design producing three distinct regimes is the validity check of the method, not a limitation. If the pipeline manufactured gaps, all three targets would fail. Two of three showing near-zero gaps is exactly what rules out a pipeline artifact and attributes the gap to the model. We present this as an existence result that the paper already states explicitly.

Presentation and clarity (Reviewers 1 and 2)

Reviewer 2 raises writing issues most directly (misleading name, forward and reverse flips used before they are defined, problem formulation that under-serves the reader). Reviewer 1 raises related clarity gaps (the three strategies are used before being defined, the auditor implementation is under-specified, the Equation 4 weights are unexplained). The consensus response is agreement plus concrete revisions. We add self-contained strategy definitions, an intuitive gloss for flips at first mention with a forward reference, a restructured Section 3 with a worked example and a notation glossary, a full auditor specification appendix, and an explicit rationale for the Equation 4 weights.

Reliance on a single model family and judge robustness (Reviewer 3, related to Reviewer 1)

Reviewer 3 flags that Gemini-2.5-Flash serves generation, gating, auditing, and judging, which risks correlated errors. Reviewer 1 independently asks for a larger human evaluation of the LLM-as-a-judge. Both point at the same underlying question of how much the findings depend on one model's judgment. Our consensus response points to two decoupling mechanisms already in the paper (the judge-independent length signal in Figure 3 and the human validation with Cohen's kappa 0.79) and then proposes a cross-judge and cross-auditor robustness experiment plus an expanded human audit above 10 percent.

Reproducibility and artifact release (Reviewer 1, endorsed toward all)

Reviewer 1 gave low scores on Reproducibility, Datasets, and Software. Reviewers 2 and 3 gave middling to good scores but would still benefit from release. The consensus response is a single commitment to a full gated artifact release (harness, 1,512 certified pairs with metadata, auditor, judge, GEPA seed, human annotations, analysis scripts), which simultaneously answers Reviewer 1's scores and supports the independent replication requested by Reviewers 2 and 3.

Novelty beyond known intuition (Reviewer 2, indirectly Reviewer 3)

Reviewer 2 argues that imperfect out-of-distribution transfer is folk wisdom. Reviewer 3's benchmark-comparison request implicitly asks what is new relative to existing multilingual safety work. Our consensus response is that the contribution is converting folk intuition into a controlled, causally-attributed, severity-graded, model-resolved measurement, and that in two of three models the intuition is empirically wrong rather than merely unquantified.

Summary of Proposed Revisions

Experiments proposed (all tagged as proposals, no fabricated numbers)

- Add a role-prompting strategy cell to the evaluation grid under the identical harness. Addresses Reviewer 1.
- Add a semantic-direct encoding strategy family (cipher, substitution, leetspeak, structured obfuscation) within the Hinglish/RH register. Addresses Reviewer 1.
- Run an Equation 4 generator-objective weight-sensitivity ablation and re-measure the headline gaps and strategy ordering. Addresses Reviewer 1.
- Expand the human judge validation to at least 160 pairs (above 10 percent) with a second annotator and inter-annotator agreement. Addresses Reviewer 1, and relevant to Reviewer 3.
- Cross-judge and cross-auditor robustness study using independent model families (for example a Claude-family judge and an open safety judge). Addresses Reviewer 3, and relevant to Reviewer 1.
- Empirical bridge that re-scores externally sourced red-teaming prompts (StrongREJECT, HarmBench behaviors, IndicJR Romanized-Hindi split) through the graded judge and, where matched, the equivalence auditor. Addresses Reviewer 3.
- English-to-unusual-English register control to separate language-conditioned failure from generic distribution shift. Addresses Reviewer 2.
- Certification-ablation comparing certified versus uncertified (drift-allowed) pairs to make the causal role of the auditor empirically visible. Addresses Reviewer 2.
- Additional open-weight model families (Llama, Mistral, Gemma, DeepSeek) on the same harness. Addresses Reviewer 3.

- At least one additional Indic language and its Romanized script through a re-parameterized language-fidelity axis. Addresses Reviewer 3, and relevant to Reviewer 2.

Writing and presentation revisions

- Retitle and rescope the abstract and Section 1 so Hinglish/RH is stated as the single evaluated instantiation and other Indic registers are a stated extension. Add one sentence stating the harness is language-agnostic apart from two language-specific checks. Addresses Reviewer 2.
- Add an intuitive one-line gloss for forward and reverse flips at first mention with a forward reference to Section 3 and Figure 1. Standardize flip terminology across the abstract and later sections. Addresses Reviewer 2.
- Restructure Section 3 to lead with a worked certified pair, add a notation glossary, state the central directional-asymmetry claim as a single up-front sentence, and tighten the link from formulation to the causal reading. Addresses Reviewer 2.
- Add self-contained definitions of symbolic masking, scenario nesting, and logical appeal in Section 4, and replace the opaque `CLs1SMCR` strategy code in Appendix B with a readable label. Addresses Reviewer 1.
- Sharpen Section 1 and the Conclusion to state explicitly what the paper adds beyond the folk intuition (controlled causal attribution, graded severity, model-resolved regimes). Addresses Reviewer 2.
- Make the single-model asymmetry read as evidence of model specificity rather than a thin result, in the abstract and Section 6.2. Signpost the existing category-by-strategy ablations (Figure 2, Section 6.1, Appendix A.3) more clearly. Addresses Reviewer 3.

Presentation and specification additions

- Add an appendix subsection with the full equivalence auditor specification (auditor prompt, output schema, per-dimension anchors, aggregation function mapping the nine dimensions to the scalar equivalence score, and the exact flag logic), plus auditor rejection statistics over the full sweep. Addresses Reviewer 1.
- Add an explicit rationale for the Equation 4 weights and a clear statement that Equation 4 governs only the frozen GEPA generator and does not relax the hard certification gate. Addresses Reviewer 1.
- Make the judge-independent length signal (Appendix A.1, Figure 3) and the human validation (Appendix A.5) more prominent as orthogonal corroboration. Addresses Reviewers 1 and 3.

Missing comparisons and positioning

- Add a compact positioning table to Related Work stating, for each external benchmark and pipeline, what quantity it measures, whether it enforces prompt-pair equivalence, and

whether it scores binary or graded. Addresses Reviewer 3.

Artifact release

- Release, under a gated research-use license, the full harness, all 1,512 certified pairs with per-pair metadata, the equivalence auditor, the graded judge, the GEPA generator seed and objective, the human annotation data, and the analysis scripts that regenerate Table 1 and Figures 2 and 3. Add a reproducibility appendix with model versions, decoding settings, and run instructions. Addresses Reviewer 1, and supports Reviewers 2 and 3.

Meta-Response Strategy

Overall tone

The tone is appreciative and confident. Two of three reviewers scored the paper below the acceptance line (Overall Assessment 2 from Reviewers 2 and 3), so the rebuttal has to move them. We do this by conceding every presentation point quickly and without defensiveness, then spending our strongest argumentation on the two substantive challenges that actually drive the low scores, which are the novelty concern (Reviewer 2) and the single-family or single-model reliance concern (Reviewer 3). Reviewer 1 is already positive (Overall Assessment 4), so for that review the strategy is to convert agreement into concrete commitments, especially on reproducibility where the numeric scores were low.

Which concerns need new experiments versus clarification only

Clarification only. Reviewer 1's Equation 4 weights (scoping the objective to the frozen generator), Reviewer 2's flip definitions and problem-formulation concerns, Reviewer 3's benchmark-comparison concern (different measured quantity), and Reviewer 3's single-model concern (a deliberate existence result) are answered primarily by clarification and revision. These cost little and should be granted fully.

New experiments. Reviewer 3's cross-judge and cross-auditor robustness study and the external-prompt bridge are the highest-value additions because they directly attack the two concerns most responsible for the low scores. Reviewer 2's English-to-English register control and certification-ablation are the sharpest possible answers to the folk-wisdom objection because they isolate language from generic distribution shift. Reviewer 1's expanded human audit, role-prompting cell, encoding strategy, and weight-sensitivity ablation are lower-risk breadth additions. Every proposed experiment is tagged as a proposal with an expected direction only, and no fabricated measured numbers appear anywhere.

Contradictions between reviewers and how to navigate them

There is no hard contradiction, but there is a tension in emphasis worth managing. Reviewer 3 treats the single-model asymmetry as a weakness, while our whole design argument treats it as the central validity signal. We navigate this by making the "same harness, three regimes" logic explicit and prominent rather than buried in Section 6.2, so the reader sees specificity as the intended result. A second tension is that Reviewer 1 wants more strategies and more human annotation (breadth and depth on the instrument) while Reviewer 2 questions whether the finding is novel at all. We reconcile these by framing every breadth addition as strengthening the measurement contribution, which is exactly the axis on which we defend novelty against Reviewer 2. A third tension is on scope. Reviewer 3 wants more languages and models, and we agree, but we also must defend the deliberate single-pair, three-model design as the right way to establish a clean controlled measurement first. We hold both by presenting breadth as the next step built on a validated primitive rather than as a gap that undermines the current result.

Strategic priorities

The single highest-leverage move is the cross-judge and cross-auditor robustness experiment, since it simultaneously answers Reviewer 3's Point 1 and reinforces the response to Reviewer 1's judge-validation request, and it is the cleanest way to show the findings are not an artifact of one model family. The second priority is the honest retitle and Section 3 restructure, which are cheap and directly lift Reviewer 2's presentation objections. The third is the full artifact release, which lifts Reviewer 1's low reproducibility scores and supports every reviewer's replication interest at once.